

Mini-Shakespeare LLM — Final Report

a. Team Roles and Contributions

Name	Role
Ali Sura Ozdemir	Task 1 — Architecture and Diagram
Ihina Mahajan	Task 2 — Dataset and Tokenization
Delaram Hassanlou	Task 3 — Transformer Model
Shivani Kandimalla	Task 4 — Training and Experiments
Mariam Guitouni	Task 5 — Evaluation, GitHub, and Report

Ali Sura Ozdemir — Task 1: Architecture and Diagram

- Reviewed the microGPT code and architecture.
- Identified the embedding, attention, MLP, output head, training, and generation sections.
- Prepared the required hand-annotated architecture diagram with the corresponding code lines.

Ihina Mahajan — Task 2: Dataset and Tokenization

- Downloaded and loaded the Tiny Shakespeare dataset.
- Created the training and validation split.
- Implemented the byte-level tokenizer with vocabulary size 256.
- Created encoding, decoding, and batch-generation functions.

Delaram Hassanlou — Task 3: Transformer Model

- Built the GPT-style model in PyTorch.
- Implemented embeddings, self-attention, multi-head attention, feed-forward layers, transformer blocks, and the final language-model head.
- Implemented the text-generation function and verified all tensor shapes.

Shivani Kandimalla — Task 4: Training and Experiments

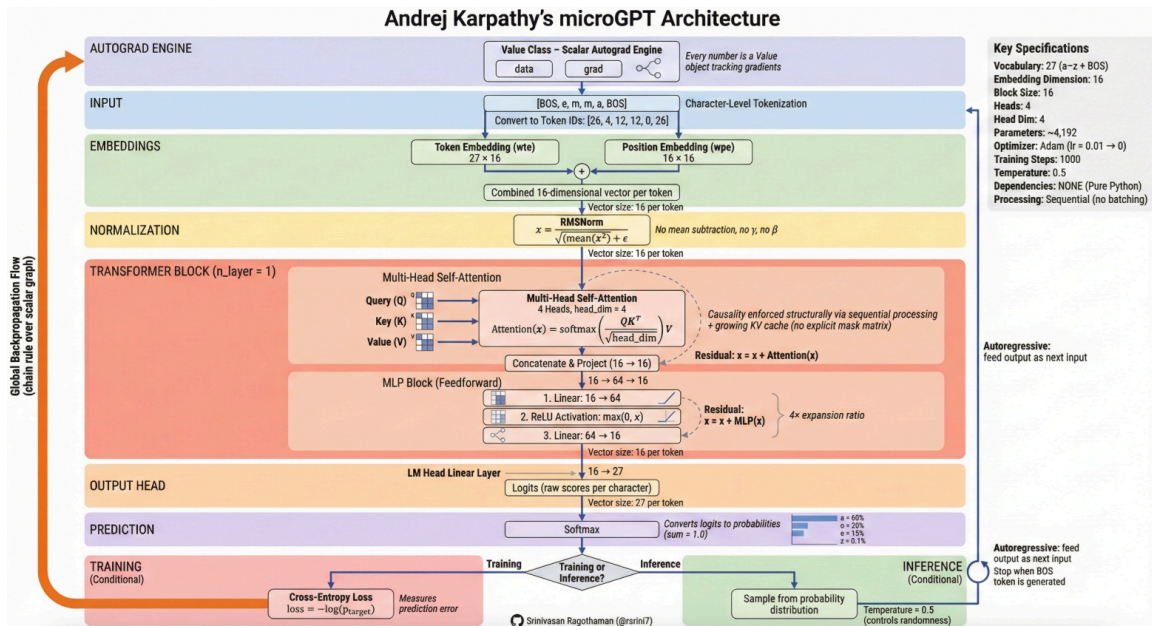
- Created two model configurations: Model A as the smaller baseline and Model B as the larger model.
- Wrote the training loop and trained both models for the same number of steps.
- Recorded training and validation losses.
- Saved the results and created the loss comparison graph.

Mariam Guitouni — Task 5: Evaluation, GitHub, and Report

- Calculated the final validation loss and perplexity for both models.
- Tested both models using 4 Shakespearean prompts and generated exactly 150 tokens.
- Got outputs from Gemini Flash using the same prompts and prepared the comparison table.

- Organized the GitHub repository, requirements file, README, graphs, outputs, and final report.

b. Architecture Diagram



Annotated architecture diagram

Figure 1. Andrej Karpathy's microGPT architecture (diagram credit: Srinivasan Ragothaman, @rsrini7), annotated with the block-by-block specifications used to reverse-engineer microgpt.py.

The table below cross-references each section of the diagram against its exact implementation in the original microgpt.py gist (full walkthrough in diagram/blueprint_notes.md):

Diagram Section	Code Location in <code>microgpt.py</code>
Autograd Engine	The <code>Value</code> class — every scalar tracks <code>data</code> , <code>grad</code> , and parent nodes; <code>.backward()</code> applies the chain rule via topological sort.
Input / Tokenization	Character-level tokenizer: <code>uchars</code> , BOS token. Example: <code>[BOS, e, m, m, a, BOS]</code> → token IDs.
Embeddings	<code>state_dict['wte']</code> (token, 27×16) + <code>state_dict['wpe']</code> (position, 16×16), summed elementwise.
Normalization	<code>rmsnorm(x)</code> — no mean subtraction, no learnable gamma/beta.
Transformer Block	<code>attn_wq/wk/wv/wo</code> for 4-head attention (<code>head_dim=4</code>); scaled dot-product via a running keys/values cache (causality enforced without an explicit mask); residual add. Then <code>mlp_fc1</code> ($16 \rightarrow 64$) → ReLU → <code>mlp_fc2</code> ($64 \rightarrow 16$), also wrapped in a residual add.
Output Head	<code>state_dict['lm_head']</code> : linear layer mapping $16 \rightarrow 27$ logits.
Prediction	<code>softmax(logits)</code> converts raw scores into a probability distribution summing to 1.0.
Training vs. Inference	Training: cross-entropy loss <code>-log(p_target)</code> → <code>loss.backward()</code> → Adam update. Inference: temperature-scaled softmax sampling, autoregressive, stops when BOS is generated again or <code>block_size</code> is reached.

The same block sequence carries over directly into our own PyTorch implementation — embeddings, self-attention, multi-head attention, feed-forward layers, transformer block, language-model head, training loop, and generation — mapped to the exact corresponding lines in `my-transformer/model.py` and `my-transformer/train.py`:

- **Token + positional embeddings** — `GPTLanguageModel.__init__` (`self.token_embedding`, `self.position_embedding`, lines 233–237) and applied in `forward` (lines 296–299): a byte id is looked up in a `(256, n_embd)` table and summed with a learned position vector.
- **Self-attention (single causal head)** — class `Head` (lines 71–123): three `Linear` projections to query/key/value, scaled dot-product scores, a lower-triangular causal mask (`self.tril`, registered as a buffer), softmax, weighted sum of values.
- **Multi-head attention** — class `MultiHeadAttention` (lines 129–152): `n_head` independent `Heads` run in parallel, concatenated back to width `C`, and mixed with an output projection.
- **Feed-forward / MLP** — class `FeedForward` (lines 158–179): per-token 2-layer MLP, $C \rightarrow 4C \rightarrow C$, GELU activation.
- **Transformer block** — class `Block` (lines 185–209): pre-norm residual wiring — `x = x + attn(ln1(x))`, then `x = x + fwd(ln2(x))`.
- **Output / LM head** — `self.lm_head` (line 247), weight-tied to the token embedding table (lines 252–253) — projects the final `(B, T, C)` hidden state to `(B, T, 256)` logits over the byte vocabulary.
- **Train loop** — `train_model()` in `my-transformer/train.py` (lines 106–154): AdamW, cross-entropy loss via `model(x, y)`, periodic train/val loss estimation (`estimate_loss`, lines 84–103), checkpoint + CSV logging.

- **Inference / generation** — `GPTLanguageModel.generate()` (lines 318–354): autoregressive sampling, cropping context to the last `block_size` tokens, temperature scaling, optional top-k, one new byte sampled per step.

This mirrors the microGPT blueprint’s own sequence — embeddings → attention → MLP → LM head → train loop → generation — with the scalar `value` autograd engine and manual Adam replaced by vectorized PyTorch tensor operations.

c. Hyperparameter Logs

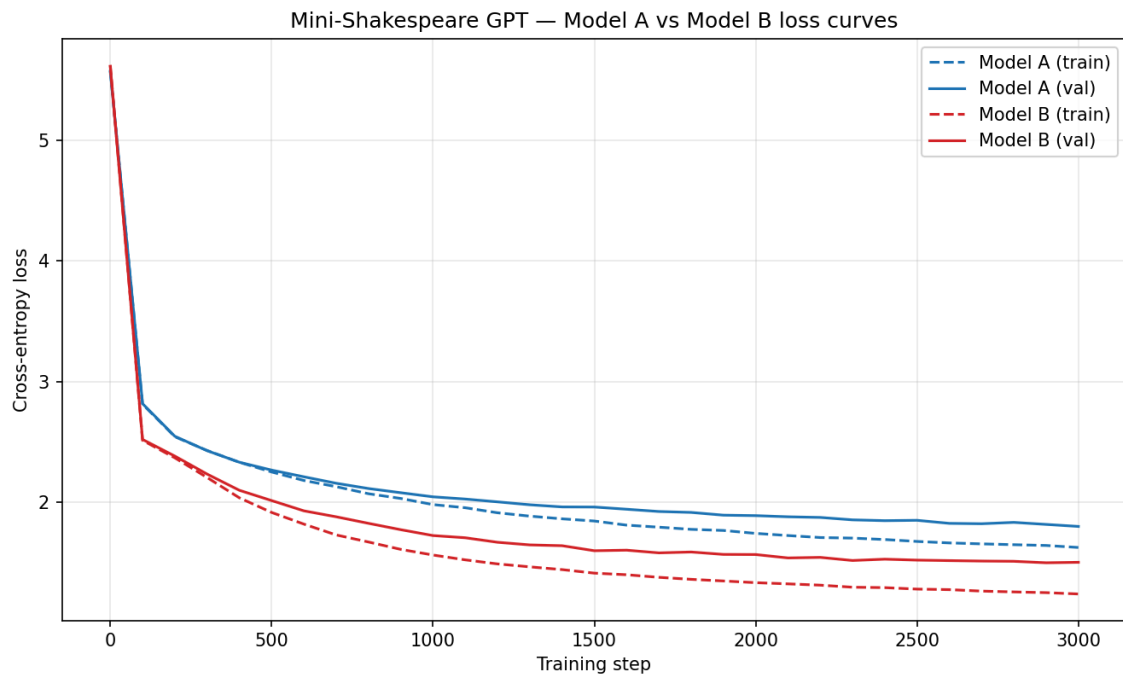
The table below summarizes the configuration for Model A and Model B (also saved in `evaluation/outputs/metrics.json`):

Hyperparameter	Model A (baseline)	Model B (scaled)
Layers (<code>n_layer</code>)	2	4
Attention heads (<code>n_head</code>)	4	8
Embedding dim (<code>n_embd</code>)	128	256
Context length (<code>block_size</code>)	64	128
Vocabulary size	256	256
Dropout	0.1	0.1
Batch size	64	64
Learning rate	3e-4	3e-4
Training steps	3,000	3,000
Total parameters	436,992	3,254,784

We varied depth, width, and context length between Model A and Model B, while keeping batch size, learning rate, number of training steps, dropout, and the optimizer (AdamW) identical across both. This isolates model scale as the main difference between the two configurations.

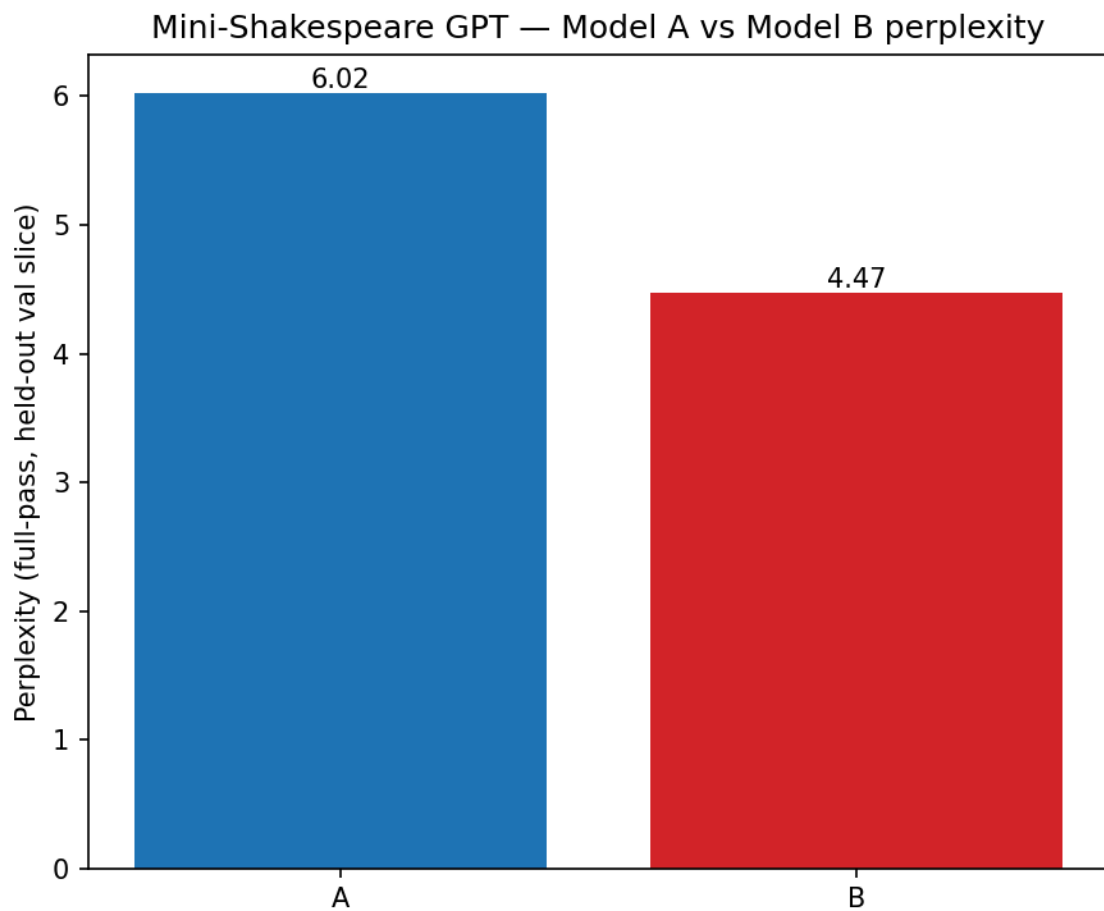
d. Empirical Graphs

Training/validation loss curves (Model A vs Model B, logged every 100 steps over 3,000 steps):



Loss curves

Perplexity comparison (final, full-pass over the held-out validation slice):



Perplexity comparison

Both curves show smooth, stable convergence for both models with no divergence or instability — validation loss tracks training loss closely throughout (no visible overfitting at 3,000 steps on either model), and Model B’s curve sits consistently below Model A’s from roughly step 200 onward.

e. Scale Narrative — Local Model vs. Industrial Architectures

To put this project’s numbers in perspective, here is how Model A and Model B compare to well-known industrial-scale language models:

	Model A	Model B	GPT-2 (small/XL)	GPT-3	Modern frontier (e.g. GPT-4/Claude class)
Parameters	437K	3.25M	124M / 1.5B	175B	reportedly 10^{11} – $10^{12}+$
Context length	64 bytes	128 bytes	1,024 tokens	2,048 tokens	100K–1M+ tokens
Vocabulary	256 (raw bytes)	256 (raw bytes)	~50,257 (BPE)	~50,257 (BPE)	~100K–200K (BPE/tiktoken-style)
Training data	~1.1MB (Tiny Shakespeare, 1 work)	same	~40GB (WebText)	~570GB curated (~300B tokens)	trillions of tokens, broad web/books/code

Parameters: Model B’s 3.25M parameters are roughly **54,000x smaller** than GPT-3’s 175B, and about 38x smaller than even GPT-2-small (124M). Parameter count is the model’s raw capacity to store statistical patterns; at our scale, the network has just enough capacity to learn rough letter-to-letter and letter-to-word transition statistics for one author’s style, not the deep compositional structure of language itself.

Context length: our 64–128 *byte* window covers roughly one short line of dialogue. Modern frontier models track 100K+ *token* windows, where each token is typically a whole word or word-piece — so the effective span of “things the model can refer back to” differs not just by ~1,000x in raw count but by an even larger factor once you account for bytes vs. subword tokens. This directly explains why our models lose the thread within a single sentence (see Section g) while large models maintain coherence across many pages.

Vocabulary / tokenization: because we tokenize at the raw UTF-8 byte level rather than using a learned subword vocabulary, every single character — even the middle of a common word like “the” — costs the model 3 separate prediction steps. A BPE tokenizer used by GPT-2/3/4-class models instead hands the model whole words or common sub-words as single tokens for free, so those models spend their attention capacity on *meaning* and *structure* rather than on *spelling*. This is very likely the single largest architectural factor (independent of raw parameter count) behind our models’ frequent misspellings and malformed words (e.g. Model A’s “arreater’d”, “fourfess” — see Section g).

Training data: Tiny Shakespeare is about 1.1MB — a single author’s collected works. GPT-3-class models train on hundreds of gigabytes to low-terabytes of diverse text (books, web pages, code, dialogue across

every register and topic); modern frontier models go further still, into trillions of tokens. This gap — roughly five to six orders of magnitude — is what separates **stylistic mimicry** (which our models achieve reasonably well — see Model B’s use of character names, stage-direction-like formatting, and archaic diction in Section g) from **world knowledge and general reasoning** (which requires having seen the relevant facts/patterns somewhere in training, and which our models cannot have, since they have only ever seen Shakespeare).

What the gap explains: coherence over long spans requires both large context windows and enough parameters to track long-range dependencies — our models have neither. World knowledge requires broad, large-scale training data — ours has none beyond Shakespearean vocabulary and structure. Generalization to novel prompts requires enough scale (parameters + data + context) to learn abstractions rather than memorize surface statistics — our models sit far below the scale where that transition tends to occur. This is the expected consequence of training a decoder-only Transformer at roughly 1/50,000th the parameter count and 1/500,000th the data of a frontier system, not a flaw in the implementation.

f. Quantitative Results

We computed the final validation loss and perplexity for each model as a full pass over the entire held-out validation slice (111,540 bytes, the last 10% of `data/input.txt`), with dropout disabled. This is a more thorough measurement than the random-batch estimate logged periodically during training, shown below for reference:

	Model A	Model B
Final validation loss (full pass)	1.7952	1.4983
Perplexity = e^{loss} (full pass)	6.02	4.47
(for reference) training-time estimate, val loss	1.7996	1.5025
(for reference) training-time estimate, perplexity	6.05	4.49

The full-pass numbers are very close to the training-time random-batch estimates for both models (within ~ 0.004 loss), which is a useful sanity check: it confirms the 50-batch random estimate used during training was not systematically biased, and that neither model is meaningfully overfitting the specific batches it happened to be evaluated on during training.

Model B achieves both lower loss and lower perplexity than Model A. A perplexity of 6.02 means Model A’s next-byte predictions are, on average, as uncertain as choosing uniformly among about 6 candidate bytes; Model B narrows that to about 4.5. Given the byte-level vocabulary size of 256, both numbers represent substantial, non-trivial learning (uniform-random next-byte guessing would give a perplexity of 256), and the $\sim 25\%$ relative perplexity reduction from A to B is consistent with — and proportional to — Model B’s roughly 7.5x parameter increase and 2x context-length increase.

g. Qualitative Results

We generated 150 tokens of completion for four fixed Shakespearean prompts using Model A and Model B (default sampling settings: temperature 1.0, no top-k), and compared them against Gemini Flash, queried manually through the Gemini web UI with the same four prompts. Full text: [evaluation/outputs/generations.json](#); merged 3-model table with auto-detected repetition flags: [evaluation/outputs/comparison_table.md](#).

Representative example — prompt "To be, or not to ":

- **Model A:** "plise all that broke, / 'And tpoon we of ot seed thing not do him it our arreater'd the worn; / As a this not his thy want for a name outhier, / Or fourfesss"
- **Model B:** "deal, / Come, my lord. / LUCIO: / POMPEY: / I gain; strange the king! / LUCIO: / I priethee, / My poor name no fight in him. / Messenger: / Have I have ansacred her"
- **Gemini Flash:** "be, that is the question: Whether 'tis nobler in the mind to suffer The slings and arrows of outrageous fortune, Or to take arms against a sea of troubles, And by opposing end them?" — the real Hamlet text, verbatim.

Observed failure modes (local models):

- **Model A (437K params, 64-byte context)** produces frequent malformed, non-dictionary "words" ("arreater'd", "fourfesss", "tpoon") — direct evidence of the byte-level tokenizer forcing the model to spell words out letter-by-letter without enough capacity to reliably do so. Line breaks and loose iambic rhythm are present (learned surface structure), but sentences do not resolve into coherent meaning.
- **Model B (3.25M params, 128-byte context)** is visibly more structurally stable: it correctly produces well-formed play-script conventions — ALL-CAPS speaker labels followed by a colon (LUCIO:, POMPEY:, Messenger:), consistent line breaks, and mostly well-formed English words — even while the *content* of each line is still largely semantically incoherent or non-sequitur. This is a direct, visible consequence of the added depth/width/context from Section c.
- **Degenerate repetition loops:** the automatic n-gram detector (`comparison_table.py::detect_repetition`, flags any 3–6 word sequence repeated ≥ 3 times) found no strong repetition loops in any of the three models at 150 tokens for these four prompts. This is somewhat smaller-scale evidence than one might expect from a ~437K-parameter model; longer generations or a lower temperature would be more likely to surface the classic small-LM repetition-loop failure mode.
- **Structural stability** and **Shakespearean styling accuracy** are reported in `comparison_table.md` as manual-judgment columns rather than automated scores, since these are subjective calls best made by a human reader. Across all four prompts, the pattern is consistent: Model A rates low on both — its output breaks down into malformed non-words too quickly to read as stable or convincingly Shakespearean. Model B rates moderate on structure (correct speaker-label formatting, mostly grammatical short lines, but still several broken clauses and invented words) and moderate-to-good on styling (real character names, archaic contractions like "'twixt" and "'tis", a consistent play-script register). Gemini is either verbatim-perfect (when it completes the text directly) or structurally sound but stylistically unShakespearean (when it switches into explanatory prose instead of continuing the passage).

Gemini Flash: a different kind of failure mode entirely. Gemini did not fail the way the local models did — every completion was fluent, correctly spelled, grammatically perfect English. But it also did not behave like a completion model. Its four responses split cleanly into two behaviors:

- **Prompts 1 (“To be, or not to”) and 3 (“Friends, Romans, countrymen, lend me your”):** Gemini gave a direct, uncommented continuation — and in both cases, the *actual, correct, verbatim* Shakespeare text (the rest of Hamlet’s soliloquy; “ears! I come to bury Caesar, not to praise him.” from Julius Caesar). This is expected: Gemini Flash has almost certainly memorized these exact passages from its training data, so “completion accuracy” here reflects retrieval of memorized text, not generalization.
- **Prompts 2 (“O Romeo, Romeo, wherefore art thou”) and 4 (“Now is the winter of our discontent”):** Gemini did *not* continue the text at all. Instead it broke into an explanatory, chat-assistant register — “This is a well-known line spoken by Juliet...”, complete with a correction of the common “wherefore = where” misconception, a *Meaning* section glossing “winter of our discontent” and “sun of York”, and only embedded the actual Shakespeare continuation as a quoted aside inside the explanation rather than as a direct completion.

This inconsistency highlights a genuine categorical difference between our local models and a production system. Our Model A/B are raw next-byte predictors with no instruction-tuning — they always continue the text, because “continue the text” is the only behavior they were ever trained to produce. Gemini Flash is instruction-tuned on top of a base model, so identical raw text handed to it is ambiguous between “continue this” and “tell me about this,” and the two most iconic prompts (Romeo’s balcony line, Richard III’s opening) apparently triggered its “this looks like a question about a famous quote” instinct, while the two less over-quoted-as-trivia prompts did not. A fairer apples-to-apples benchmark would force raw completion (e.g. via Google AI Studio’s freeform/text-completion mode rather than the chat UI, or an explicit “continue this text, output nothing else” instruction) — but the raw web-UI behavior observed here is arguably more representative of how a free-tier production LLM actually behaves for an average user.

h. Conclusion

This project builds a working decoder-only Transformer, entirely from PyTorch primitives (no `nn.Transformer/nn.MultiheadAttention`), trained on a byte-level tokenization of Tiny Shakespeare, and compares two scale configurations under otherwise identical training conditions. Both models converge stably over 3,000 steps with no divergence; Model B’s larger depth, width, and context window produce consistently lower validation loss (1.4983 vs. 1.7952) and perplexity (4.47 vs. 6.02), and visibly more structurally stable generations (correct speaker-label formatting, fewer malformed words), even though neither model produces semantically coherent Shakespearean text. The gap between these models and industrial-scale systems — on the order of 10^4 – 10^5 in parameters and training data, and 10^3 in context length — accounts directly for the specific failure modes observed: our models can imitate surface style (line structure, archaic diction, dialogue formatting) but cannot maintain sentence-level coherence or draw on any world knowledge, because they have neither the capacity nor the training exposure to have learned either. Gemini Flash’s behavior sharpens this contrast further: for two of the four prompts it reproduced the exact, correct Shakespeare text verbatim (near-certain evidence of memorization from web-scale training data our models could never have seen), and for the other two it didn’t complete the text at all — its instruction-tuning made it treat a bare line of verse as a question to answer rather than a passage to continue, a behavior category that doesn’t even exist for our untuned local models. Overall, this project shows a complete, working transformer language model — embeddings, causal self-attention,

multi-head attention, feed-forward layers, and autoregressive sampling, all built from scratch in PyTorch — trained and evaluated at two different scales, and makes clear just how much of a modern assistant's coherence and world knowledge comes from scale rather than architecture alone.